



**Generation Interconnection
Impact Study Report
for
Queue Project AE2-255
MOLECREEK 345 KV
25 MW Capacity / 100 MW Energy**

February, 2020

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1 Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an Interconnection Customer may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The project developer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

The Interconnection Customer seeking to interconnect a wind or solar generation facility shall maintain meteorological data facilities as well as provide that meteorological data which is required per Schedule H to the Interconnection Service Agreement and Section 8 of Manual 14D.

An Interconnection Customer with a proposed new Customer Facility that has a Maximum Facility Output equal to or greater than 100 MW shall install and maintain, at its expense, phasor measurement units (PMUs). See Section 8.5.3 of Appendix 2 to the Interconnection Service Agreement as well as section 4.3 of PJM Manual 14D for additional information.

2 General

The Interconnection Customer (IC), has proposed a Wind generating facility located in Woodford County, Illinois. This is an uprate of the proposed AE1-113 queue request, and the installed facilities will have a total capability of 400 MW, an increase of 100 MW with 25 MW of this output being recognized by PJM as Capacity. The proposed in-service date for this project is December 31, 2021. This study does not imply a TO commitment to this in-service date.

Queue Number	AE2-255
Project Name	MOLECREEK 345 KV
Interconnection Customer	TGE Illinois 181, LLC
State	Illinois
County	Woodford
Transmission Owner	ComEd
MFO	400
MWE	100
MWC	25
Fuel	Wind
Basecase Study Year	2022

2.1 Point of Interconnection

AE2-255 will interconnect with the ComEd transmission system as an uprate to AE1-113 at the Mole Creek 345kV substation.

2.2 Cost Summary

The AE2-255 project will be responsible for the following costs associated with the physical interconnection of the project:

Description	Total Cost
Attachment Facilities	\$200,000
Direct Connection Network Upgrade	\$0
Non Direct Connection Network Upgrades	\$0
Total Costs	\$200,000

Notes on Cost Estimates:

1. These estimates are Order-of-Magnitude estimates of the costs that ComEd would bill to the customer for this interconnection. These estimates are based on a one-line electrical diagram of the project and the information provided by the IC.
2. There were no site visits performed for these estimates. There may be costs related to specific site related issues that are not identified in these estimates. The site reviews will be performed during the Facilities Study or during detailed engineering.
3. These estimates are not a guarantee of the maximum amount payable by the IC and the actual costs of ComEd's work may differ significantly from these estimates. The IC will be responsible for paying actual costs of ComEd's work in accordance with Sections 212.1 and 217 of the PJM Open Access Transmission Tariff.
4. The IC is responsible for all engineering, procurement, testing and construction of all equipment on the IC's side of the POI.
5. These cost estimates do not include cost of acquiring right-of-way for the transmission line and purchasing any additional land, if needed, for the line terminations. The need and cost of acquiring property and associated legal costs will be investigated during Facilities Study for this project.

In addition, the AE2-255 project may be responsible for a contribution to the following costs associated with network upgrades:

Description	Total Cost
System Upgrades	\$7,347,400

Cost allocations for these upgrades will be provided in Section 11.5.

3 Transmission Owner Scope of Work

To accommodate interconnection of AE2-255, the relaying, SCADA, Communication and metering between the proposed AE1-113 wind farm and TSS 908 Mole Creek would be reviewed and upgraded if needed.

4 Attachment Facilities

The total preliminary cost estimate for the Attachment work is given in the table below. These costs do not include CIAC Tax Gross-up.

Description	Total Cost
Review and upgrade the relaying, SCADA, communication and metering between AE1-113 and TSS Mole Creek.	\$200,000
Total Attachment Facility Costs	\$200,000

5 Direct Connection Cost Estimate

The total preliminary cost estimate for the Direct Connection work is given in the table below. These costs do not include CIAC Tax Gross-up.

None.

6 Non-Direct Connection Cost Estimate

The total preliminary cost estimate for the Non-Direct Connection work is given in the table below. These costs do not include CIAC Tax Gross-up.

None.

7 Schedule

ComEd would take approximately 18-months to review and possibly upgrade SCADA, Communication, relays and metering after the ISA / ICSA are signed.

8 Interconnection Customer Requirements

ComEd interconnection requirements can be found at <https://www.pjm.com/planning/design-engineering/to-tech-standards/private-comed.aspx>

To the extent that these Applicable Technical Requirements and Standards may conflict with the terms and conditions of the Tariff, the Tariff shall control.

ComEd distribution line drops to move customer cranes and heavy equipment is not part of PJM process. The customer should directly contact ComEd New Business Group to arrange for line drops, if needed.

9 Revenue Metering and SCADA Requirements

9.1 PJM Requirements

The Interconnection Customer will be required to install equipment necessary to provide Revenue Metering (KWH, KVARH) and real time data (KW, KVAR) for IC's generating Resource. See PJM Manuals M-01 and M-14D, and PJM Tariff Section 8 of Attachment O.

9.1.1 Meteorological Data Reporting Requirement

The solar generation facility shall provide the Transmission Provider with site-specific meteorological data including:

- Temperature (degrees Fahrenheit)
- Atmospheric pressure (hectopascals)
- Irradiance
- Forced outage data

10 Network Impacts

The Queue Project AE2-255 was evaluated as a 100.0 MW (Capacity 25.0 MW) injection as an uprate to AE1-113 at the Mole Creek 345kV substation in the ComEd area. Project AE2-255 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AE2-255 was studied with a commercial probability of 100%. Potential network impacts were as follows:

11 Summer Peak Load Flow

11.1 Generation Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

None

11.2 Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CK T ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADIN G %	POST PROJECT LOADIN G %	AC D C	MW IMPAC T
8900984	930750	AB1-122 TAP	345.0	CE	270716	DRESDEN ; B	345.0	CE	1	COMED_P7_345-L0301__B-S+_345-L9801__R-S_FSA	tower	1768.0	98.13	101.24	DC	55.0

11.3 Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CK T ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADIN G %	POST PROJECT LOADIN G %	AC D C	MW IMPAC T
8899933	270716	DRESDEN ; B	345.0	CE	275179	DRESDEN ;1M	138.0	CE	1	COMED_P4_012-45-BT5-6__	breaker	442.0	130.74	133.54	AC	12.2
8900990	270716	DRESDEN ; B	345.0	CE	274702	KENDALL ;BU	345.0	CE	1	COMED_P7_345-L11620_B-S+_345-L11622_R-S	tower	1195.0	112.52	113.72	DC	16.69
8900980	270855	POWER ON ; R	345.0	CE	349662	7TAZEWE LL	345.0	AMI L	1	COMED_P7_345-L19601_B-S+_345-L9801__R-S_FSA	tower	1479.0	131.33	134.31	DC	44.12
8900982	270855	POWER ON ; R	345.0	CE	349662	7TAZEWE LL	345.0	AMI L	1	COMED_P7_345-L0301__B-S+_345-L9801__R-S_FSA	tower	1479.0	120.1	123.08	DC	44.14
8899930	270926	WILTON ; B	345.0	CE	275232	WILTON ;3M	345.0	CE	1	COMED_P4_112-65-BT5-6__	breaker	1379.0	148.48	148.72	AC	14.81
8899939	270927	WILTON ; R	345.0	CE	275233	WILTON ;4M	345.0	CE	1	COMED_P4_112-65-BT2-3__	breaker	1379.0	151.55	151.79	AC	15.13
8899780	274702	KENDALL ;BU	345.0	CE	270810	LOCKPORT ; B	345.0	CE	1	COMED_P2-2_111_EJ-345B_2	bus	1479.0	107.33	108.5	DC	17.25
8900109	274702	KENDALL ;BU	345.0	CE	270810	LOCKPORT ; B	345.0	CE	1	COMED_P4_111-45-L16704T	breaker	1479.0	107.65	108.81	DC	17.24

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CK T ID	CONT NAME	Type	Rating MVA	PRE PROJECTIONS T LOADING %	POST PROJECTIONS T LOADING %	AC DC	MW IMPACT
8900110	274702	KENDALL;BU	345.0	CE	270810	LOCKPORT; B	345.0	CE	1	COMED_P4_111-45-L11126_	breaker	1479.0	107.39	108.56	DC	17.25
8900111	274702	KENDALL;BU	345.0	CE	270810	LOCKPORT; B	345.0	CE	1	COMED_P4_111-45-L14321T	breaker	1479.0	107.33	108.5	DC	17.25
7290249	274804	UNIV PK N;RP	345.0	CE	243229	05OLIVE	345.0	AEP	1	AEP_P4_#2978_05DUMONT 765_B	breaker	971.0	130.82	131.36	AC	9.75
7290250	274804	UNIV PK N;RP	345.0	CE	243229	05OLIVE	345.0	AEP	1	COMED_P4_023-65-BT2-3_	breaker	971.0	129.86	130.42	AC	9.82
7290251	274804	UNIV PK N;RP	345.0	CE	243229	05OLIVE	345.0	AEP	1	COMED_P4_112-65-BT4-5_	breaker	971.0	129.84	130.4	AC	9.82
7290252	274804	UNIV PK N;RP	345.0	CE	243229	05OLIVE	345.0	AEP	1	COMED_P4_112-65-BT3-4_	breaker	971.0	129.84	130.4	AC	9.82
7290253	274804	UNIV PK N;RP	345.0	CE	243229	05OLIVE	345.0	AEP	1	COMED_P4_023-65-BT4-5_	breaker	971.0	129.83	130.38	AC	9.82
8899941	275179	DRESDEN;1M	138.0	CE	271337	DRESDEN; R	138.0	CE	1	COMED_P4_012-45-BT5-6_	breaker	442.0	128.05	130.76	AC	12.2
8899932	275232	WILTON;3M	345.0	CE	270644	WILTON;	765.0	CE	1	COMED_P4_112-65-BT5-6_	breaker	1379.0	148.37	148.61	AC	14.81
8899938	275233	WILTON;4M	345.0	CE	270644	WILTON;	765.0	CE	1	COMED_P4_112-65-BT2-3_	breaker	1379.0	151.43	151.67	AC	15.13
8900942	930740	AB1-122 TAP	345.0	CE	270717	DRESDEN; R	345.0	CE	1	COMED_P7_345-L19601_B-S_+ 345-L9801_R-S_FSA	tower	1195.0	133.98	134.88	AC	12.15
8900943	930740	AB1-122 TAP	345.0	CE	270717	DRESDEN; R	345.0	CE	1	COMED_P7_345-L0301_B-S_+ 345-L9801_R-S_FSA	tower	1195.0	130.98	132.0	DC	12.16
8900983	930750	AB1-122 TAP	345.0	CE	270716	DRESDEN; B	345.0	CE	1	COMED_P7_345-L19601_B-S_+ 345-L9801_R-S_FSA	tower	1768.0	101.33	104.44	DC	55.0

11.4 Potential Congestion due to Local Energy Deliverability

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Note: Only the most severely overloaded conditions are listed below. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed which shall study all overload conditions associated with the overloaded element(s) identified.

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CK T ID	CONT NAME	Type	Rating MVA	PRE PROJECTIONS T LOADING %	POST PROJECTIONS T LOADING %	AC DC	MW IMPACT
7290972	255113	17STILLWELL	345.0	NIPS	243219	05DUMONT	345.0	AEP	1	AEP_P1-2_#695A	operation	1409.0	169.18	169.36	AC	14.27
8900682	270716	DRESDEN; B	345.0	CE	275179	DRESDEN;1M	138.0	CE	1	COMED_P1-2_345-L10805_B-S	operation	442.0	107.64	108.81	AC	5.99
8900502	270855	POWERTON; R	345.0	CE	349662	7TAZEWELL	345.0	AMILL	1	COMED_P1-2_345-L1202_B-S-A	operation	1479.0	119.73	123.93	AC	62.14

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CK T ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADIN G %	POST PROJECT LOADIN G %	AC DC	MW IMPACT
8900668	274702	KENDALL ;BU	345.0	CE	270810	LOCKPORT ; B	345.0	CE	1	COMED_P1-2_345-L1221__B-S	operation	1479.0	106.48	107.66	DC	17.39
8900670	274702	KENDALL ;BU	345.0	CE	270810	LOCKPORT ; B	345.0	CE	1	Base Case	operation	1201.0	106.21	107.32	DC	13.33
7291129	274804	UNIV PK N;RP	345.0	CE	243229	05OLIVE	345.0	AEP	1	AEP_P1-2_#695A	operation	971.0	129.83	130.38	AC	9.82
8900692	275179	DRESDEN ;1M	138.0	CE	271337	DRESDEN ; R	138.0	CE	1	COMED_P1-2_345-L10805__B-S	operation	442.0	105.62	106.78	AC	5.99
8900447	930740	AB1-122 TAP	345.0	CE	270717	DRESDEN ; R	345.0	CE	1	COMED_P1-2_345-L1202__B-S-A	operation	1195.0	122.43	123.87	AC	18.6
8900490	930750	AB1-122 TAP	345.0	CE	270716	DRESDEN ; B	345.0	CE	1	Base Case	operation	1201.0	109.91	114.09	DC	50.1
8900491	930750	AB1-122 TAP	345.0	CE	270716	DRESDEN ; B	345.0	CE	1	COMED_P1-2_345-L0304__R-S	operation	1479.0	108.41	112.61	AC	62.26

11.5 System Reinforcements

ID	Index	Facility	Upgrade Description	Cost	Cost Allocated to AE2-255	Upgrade Number																																																																																																																
8900943, 8900942	12	AB1-122 TAP 345.0 kV - DRESDEN ; R 345.0 kV Ckt 1	ALDR rating is 1477 MVA. No upgrade is required.	\$0	\$0	N/A																																																																																																																
8900984, 8900983	1	AB1-122 TAP 345.0 kV - DRESDEN ; B 345.0 kV Ckt 1	ALDR rating is 2033 MVA. No upgrade is required.	\$0	\$0	N/A																																																																																																																
8899930, 8899932, 8899939, 8899938	5, 6, 10, 11	WILTON ;3M 345.0 kV - WILTON ; 765.0 kV Ckt 1 & WILTON ; B 345.0 kV - WILTON ;3M 345.0 kV Ckt 1 & WILTON ; R 345.0 kV - WILTON ;4M 345.0 kV Ckt 1 & WILTON ;4M 345.0 kV - WILTON ; 765.0 kV Ckt 1	N5145: Upgrade: Build out the Wilton 765kV bus thereby allowing for 765kV L11216 (currently on Bus 6) to be relocated to Bus 8. Along with this line relocation, installation of 2-765kV BT CB's (6-8 & 8-2). This will eliminate the stuck breaker contingencies '112-65-BT4-5__' and '112-65-BT5-6__'. Project Type : CON Cost : \$ 12,000,000 Time Estimate : 30 Months The cost allocation table is as follows: <table><thead><tr><th>Queue</th><th>MW contribution</th><th>Percentage of Cost</th><th>Cost (M)</th></tr></thead><tbody><tr><td>AD1-039</td><td>18.0</td><td>1.92%</td><td>\$ 0.230</td></tr><tr><td>AD1-098</td><td>14.9</td><td>1.59%</td><td>\$ 0.191</td></tr><tr><td>AD1-100</td><td>173.4</td><td>18.49%</td><td>\$ 2.218</td></tr><tr><td>AD1-133</td><td>47.1</td><td>5.02%</td><td>\$ 0.603</td></tr><tr><td>AD2-038</td><td>22.8</td><td>2.43%</td><td>\$ 0.292</td></tr><tr><td>AD2-047</td><td>34.3</td><td>3.65%</td><td>\$ 0.438</td></tr><tr><td>AD2-066</td><td>17.7</td><td>1.89%</td><td>\$ 0.227</td></tr><tr><td>AD2-102</td><td>29.7</td><td>3.16%</td><td>\$ 0.379</td></tr><tr><td>AD2-134</td><td>17.6</td><td>1.88%</td><td>\$ 0.225</td></tr><tr><td>AD2-194</td><td>19.7</td><td>2.10%</td><td>\$ 0.252</td></tr><tr><td>AE1-070</td><td>21.49</td><td>2.29%</td><td>\$ 0.275</td></tr><tr><td>AE1-113</td><td>44.48</td><td>4.74%</td><td>\$ 0.569</td></tr><tr><td>AE1-114</td><td>21.21</td><td>2.26%</td><td>\$ 0.271</td></tr><tr><td>AE1-163</td><td>51.67</td><td>5.51%</td><td>\$ 0.661</td></tr><tr><td>AE1-166</td><td>27.91</td><td>2.98%</td><td>\$ 0.357</td></tr><tr><td>AE1-172</td><td>45.93</td><td>4.90%</td><td>\$ 0.588</td></tr><tr><td>AE1-193</td><td>45.82</td><td>4.88%</td><td>\$ 0.586</td></tr><tr><td>AE1-194</td><td>45.82</td><td>4.88%</td><td>\$ 0.586</td></tr><tr><td>AE1-195</td><td>45.82</td><td>4.88%</td><td>\$ 0.586</td></tr><tr><td>AE1-205</td><td>29.67</td><td>3.16%</td><td>\$ 0.380</td></tr><tr><td>AE1-252</td><td>27.02</td><td>2.88%</td><td>\$ 0.346</td></tr><tr><td>AE2-049</td><td>19.6</td><td>2.09%</td><td>\$ 0.251</td></tr><tr><td>AE2-107</td><td>15.2</td><td>1.62%</td><td>\$ 0.194</td></tr><tr><td>AE2-152</td><td>28.4</td><td>3.03%</td><td>\$ 0.363</td></tr><tr><td>AE2-153</td><td>35.1</td><td>3.74%</td><td>\$ 0.449</td></tr><tr><td>AE2-223</td><td>22.6</td><td>2.41%</td><td>\$ 0.289</td></tr><tr><td>AE2-255</td><td>15.1</td><td>1.61%</td><td>\$ 0.193</td></tr></tbody></table>	Queue	MW contribution	Percentage of Cost	Cost (M)	AD1-039	18.0	1.92%	\$ 0.230	AD1-098	14.9	1.59%	\$ 0.191	AD1-100	173.4	18.49%	\$ 2.218	AD1-133	47.1	5.02%	\$ 0.603	AD2-038	22.8	2.43%	\$ 0.292	AD2-047	34.3	3.65%	\$ 0.438	AD2-066	17.7	1.89%	\$ 0.227	AD2-102	29.7	3.16%	\$ 0.379	AD2-134	17.6	1.88%	\$ 0.225	AD2-194	19.7	2.10%	\$ 0.252	AE1-070	21.49	2.29%	\$ 0.275	AE1-113	44.48	4.74%	\$ 0.569	AE1-114	21.21	2.26%	\$ 0.271	AE1-163	51.67	5.51%	\$ 0.661	AE1-166	27.91	2.98%	\$ 0.357	AE1-172	45.93	4.90%	\$ 0.588	AE1-193	45.82	4.88%	\$ 0.586	AE1-194	45.82	4.88%	\$ 0.586	AE1-195	45.82	4.88%	\$ 0.586	AE1-205	29.67	3.16%	\$ 0.380	AE1-252	27.02	2.88%	\$ 0.346	AE2-049	19.6	2.09%	\$ 0.251	AE2-107	15.2	1.62%	\$ 0.194	AE2-152	28.4	3.03%	\$ 0.363	AE2-153	35.1	3.74%	\$ 0.449	AE2-223	22.6	2.41%	\$ 0.289	AE2-255	15.1	1.61%	\$ 0.193	\$12,000,000	\$193,000	N5145
Queue	MW contribution	Percentage of Cost	Cost (M)																																																																																																																			
AD1-039	18.0	1.92%	\$ 0.230																																																																																																																			
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AD2-038	22.8	2.43%	\$ 0.292																																																																																																																			
AD2-047	34.3	3.65%	\$ 0.438																																																																																																																			
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AE1-163	51.67	5.51%	\$ 0.661																																																																																																																			
AE1-166	27.91	2.98%	\$ 0.357																																																																																																																			
AE1-172	45.93	4.90%	\$ 0.588																																																																																																																			
AE1-193	45.82	4.88%	\$ 0.586																																																																																																																			
AE1-194	45.82	4.88%	\$ 0.586																																																																																																																			
AE1-195	45.82	4.88%	\$ 0.586																																																																																																																			
AE1-205	29.67	3.16%	\$ 0.380																																																																																																																			
AE1-252	27.02	2.88%	\$ 0.346																																																																																																																			
AE2-049	19.6	2.09%	\$ 0.251																																																																																																																			
AE2-107	15.2	1.62%	\$ 0.194																																																																																																																			
AE2-152	28.4	3.03%	\$ 0.363																																																																																																																			
AE2-153	35.1	3.74%	\$ 0.449																																																																																																																			
AE2-223	22.6	2.41%	\$ 0.289																																																																																																																			
AE2-255	15.1	1.61%	\$ 0.193																																																																																																																			

ID	Index	Facility	Upgrade Description	Cost	Cost Allocated to AE2-255	Upgrade Number												
8900980, 8900982	4	POWERTON ; R 345.0 kV - 7TAZEWELL 345.0 kV Ckt 1	CE N6280: The upgrade will be to mitigate the sag on the line. Upon completion of this upgrade the new ratings will be 1461/1656/1909/1912/2199 MVA (SN/SLTE/SSTE/SLD/ALDR). Project Type : FAC Cost : \$ 9,100,000 Time Estimate : 30 Months <table><tr><th>Queue</th><th>MW contribution</th><th>Percentage of Cost</th><th>Cost (M)</th></tr><tr><td>AE1-163</td><td>46.6</td><td>51.39%</td><td>\$4.6761</td></tr><tr><td>AE2-255</td><td>44.12</td><td>48.61%</td><td>\$4.4239</td></tr></table> MISO N6280: Ameren limit is 1512 MVA SE due to 0.64 miles of 2 conductor 12777 ACAR. Would need to reconductor or rebuild 0.64 miles with 2x954 Kcmil ACSR 45/7 RAIL and replace bus conductor Project Type : FAC Cost : \$ 750,000 Time Estimate : 24 Months This constraint is driven by a prior queue cycle. Per PJM cost allocation rules, Queue Project AE2-255 presently does not receive cost allocation for this upgrade. Note 1: As changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc., Queue Project AE2-255 could receive cost allocation. Note 2: Although Queue Project AE2-255 may not have cost responsibility for this upgrade, Queue Project AE2-255 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AE2-255 comes into service prior to completion of the upgrade, Queue Project AE2-255 will need an interim study.	Queue	MW contribution	Percentage of Cost	Cost (M)	AE1-163	46.6	51.39%	\$4.6761	AE2-255	44.12	48.61%	\$4.4239	\$9,100,000	\$4,423,900	N6280
Queue	MW contribution	Percentage of Cost	Cost (M)															
AE1-163	46.6	51.39%	\$4.6761															
AE2-255	44.12	48.61%	\$4.4239															

7290252, 7290253, 7290250, 7290251, 7290249	8	UNIV PK N;RP 345.0 kV - 05OLIVE 345.0 kV Ckt 1	AEP N4057: A sag check will be required for the ACSR/PE ~ 1414 ~ 62/19 ~ Conductor Section 1 to determine if the line section can be operated above its emergency rating of 971 MVA. The results could prove that no additional upgrades are necessary, that some upgrades on the circuit are necessary, or that the entire 40.61 mile section of line would need to be rebuilt. Estimated Cost: \$162,440 (2016 dollars). If deemed necessary to rebuild the entire 40.61 miles of the section of the line, Estimated Cost: \$81,220,000. Schedule: (1) Sag Study: 6 to 12 months. (2) Rebuild: The standard time required for construction differs from state to state. Project Type : FAC Cost : \$ 162,440 Time Estimate : 24-36 Months New Ratings: Rate A: 971 MVA Rate B: 1304 MVA This constraint is driven by a prior queue cycle. Per PJM cost allocation rules, Queue Project AE2-255 presently does not receive cost allocation for this upgrade. Note 1: As changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc., Queue Project AE2-255 could receive cost allocation. Note 2: Although Queue Project AE2-255 may not have cost responsibility for this upgrade, Queue Project AE2-255 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AE2-255 comes into service prior to completion of the upgrade, Queue Project AE2-255 will need an interim study. ComEd end SSTE rating is 1134 MVA. N6281.1: AEP owns limit on L97008. CE has a sag limit on 2 types of conductors as well. CE would need to address sag limit of 2 types of conductors to reach required rating. A preliminary estimate for sag mitigation is \$21.4M with an estimated construction timeline of 30 months. Upon completion of the sag mitigation the new ratings will become 1091/1399/1483/1674 MVA SN/SLTE/SSTE/SLD. Project Type : FAC Cost : \$ 21,400,000 Time Estimate : 30 Months New Ratings: Rate A: 1091 MVA Rate B: 1399 MVA Rate C: 1674 MVA Queue Project AE2-255 presently does not receive cost allocation for this upgrade as it does not meet PJM cost allocation thresholds. Note 1: As changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc., Queue Project AE2-255 could receive cost allocation. Note 2: Although Queue Project AE2-255 may not have cost responsibility for this upgrade, Queue Project AE2-255 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AE2-255 comes into service prior to completion of the upgrade, Queue Project AE2-255 will need an interim study.	\$21,562,440	\$0	N4057 N6281.1
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ID	Index	Facility	Upgrade Description	Cost	Cost Allocated to AE2-255	Upgrade Number																
8899933, 8899941	2, 9	DRESDEN ; B 345.0 kV - DRESDEN ;1M 138.0 kV Ckt 1 & DRESDEN ;1M 138.0 kV - DRESDEN ; R 138.0 kV Ckt 1	N6599: ComEd Tr. 81 at Dresden Station SSTE rating is 520 MVA. The upgrade will be installation of a second auto transformer on 345kV Blue Bus. This second transformer will require expansion of the 138kV at Dresden. Estimated construction timeline of 5-6 years subject to outage coordination at Dresden Station. Upon completion of the upgrade the ratings will remain as shown however the addition of a second auto on the Blue bus will distribute the post contingency flows. Project Type : CON Cost : \$ 65,000,000 Time Estimate : 5-6 Years During the Facilities Study, it will need to be investigated and determined if resolving the stuck breaker contingency causing the overload is a feasible and more economical solution ('COMED_P4_012-45-BT5-6__') such as adding a 345 kV breaker to the ring bus or re-terminating some line connections. This would also not increase fault levels at Dresden as a new transformer would. Queue Project AE2-255 presently does not receive cost allocation for this upgrade. Note 1: as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc., Queue Project AE2-255 could receive cost allocation. Note 2: Although Queue Project AE2-255 may not have cost responsibility for this upgrade, Queue Project AE2-255 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AE2-255 comes into service prior to completion of the upgrade, Queue Project AE2-255 will need an interim study. The potential cost allocation table is as follows: <table><tr><th>Queue</th><th>MW contribution</th><th>Percentage of Cost</th><th>Cost</th></tr><tr><td>AD1-039</td><td>12.601</td><td>26.09%</td><td>\$16,957,867</td></tr><tr><td>AE1-113</td><td>36.1</td><td>74.74%</td><td>\$48,581,781</td></tr><tr><td>AE2-255</td><td>12.2</td><td>25.26%</td><td>\$16,418,219</td></tr></table>	Queue	MW contribution	Percentage of Cost	Cost	AD1-039	12.601	26.09%	\$16,957,867	AE1-113	36.1	74.74%	\$48,581,781	AE2-255	12.2	25.26%	\$16,418,219	\$65,000,000	\$0	N6599
Queue	MW contribution	Percentage of Cost	Cost																			
AD1-039	12.601	26.09%	\$16,957,867																			
AE1-113	36.1	74.74%	\$48,581,781																			
AE2-255	12.2	25.26%	\$16,418,219																			

ID	Index	Facility	Upgrade Description	Cost	Cost Allocated to AE2-255	Upgrade Number																																
8900110, 8900111, 8900109, 8899780	7	KENDALL ;BU 345.0 kV - LOCKPORT ; B 345.0 kV Ckt 1	ComEd SSTE rating is 1568 MVA N5144: Re-conductor 16.1 miles with 2156 kcmil ACSR (84/19) Str Bluebird. Estimate is \$38.672 M with a construction time of 30 months. New ratings to be 1334/1726/1837/2084 MVA SN/SLTE/SSTE/SLD. PJM Network Upgrade N5144. Project Type : FAC Cost : \$38.672 Million Time Estimate : 36 months New Ratings: Rate A: 1334 MVA Rate B: 1726 MVA Rate C: 2084 MVA The cost allocation table is as follows: <table><tr><th>Queue</th><th>MW contribution</th><th>Percentage of Cost</th><th>Cost (M)</th></tr><tr><td>AB1-122</td><td>56.6</td><td>23.18%</td><td>\$8.9644</td></tr><tr><td>AC1-204</td><td>95.9</td><td>39.28%</td><td>\$15.1888</td></tr><tr><td>AD1-039</td><td>13.5</td><td>5.53%</td><td>\$2.1381</td></tr><tr><td>AD2-066</td><td>15.02</td><td>6.15%</td><td>\$2.3789</td></tr><tr><td>AD2-194</td><td>22.3</td><td>9.13%</td><td>\$3.5319</td></tr><tr><td>AE1-113</td><td>23.61</td><td>9.67%</td><td>\$3.7394</td></tr><tr><td>AE2-255</td><td>17.24</td><td>7.06%</td><td>\$2.7305</td></tr></table>	Queue	MW contribution	Percentage of Cost	Cost (M)	AB1-122	56.6	23.18%	\$8.9644	AC1-204	95.9	39.28%	\$15.1888	AD1-039	13.5	5.53%	\$2.1381	AD2-066	15.02	6.15%	\$2.3789	AD2-194	22.3	9.13%	\$3.5319	AE1-113	23.61	9.67%	\$3.7394	AE2-255	17.24	7.06%	\$2.7305	\$38,672,000	\$2,730,500	N5144
Queue	MW contribution	Percentage of Cost	Cost (M)																																			
AB1-122	56.6	23.18%	\$8.9644																																			
AC1-204	95.9	39.28%	\$15.1888																																			
AD1-039	13.5	5.53%	\$2.1381																																			
AD2-066	15.02	6.15%	\$2.3789																																			
AD2-194	22.3	9.13%	\$3.5319																																			
AE1-113	23.61	9.67%	\$3.7394																																			
AE2-255	17.24	7.06%	\$2.7305																																			
8900990	3	DRESDEN ; B 345.0 kV - KENDALL ;BU 345.0 kV Ckt 1	ComEd 345kV L93505 SLD & ALDR ratings are a minimum of 1284 MVA & 1477 MVA. No upgrade is required.	\$0	\$0	N/A																																
			TOTAL COST	\$137,234,440	\$7,347,400																																	

11.6 Flow Gate Details

The following indices contain additional information about each flowgate presented in the body of the report. For each appendix, a description of the flowgate and its contingency was included for convenience. However, the intent of the appendix section is to provide more information on which projects/generators have contributions to the flowgate in question. Although this information is not used "as is" for cost allocation purposes, it can be used to gage other generators impact. It should be noted the generator contributions presented in the appendices sections are full contributions, whereas in the body of the report, those contributions take into consideration the commercial probability of each project.

11.6.1 Index 1

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
8900983	930750	AB1-122 TAP	CE	270716	DRESDEN ; B	CE	1	COMED_P7_345-L19601_B-S+_345-L9801_R-S_FSA	tower	1768.0	101.33	104.44	DC	55.0

Bus #	Bus	MW Impact
274677	POWERTON ;5U	26.6918
274678	POWERTON ;6U	26.5095
274861	TOP CROP ;1U	0.7078
274862	TOP CROP ;2U	1.3739
274879	MINONK ;1U	2.9133
290021	O50 E	87.9920
293644	O22 E1	21.3776
293645	O22 E2	41.4976
918051	AA1-018 C	0.6719
918052	AA1-018 E	33.9535
930751	AB1-122 2O1	521.0075
934111	AD1-039 2	51.0587
934871	AD1-116 C	1.9774
934872	AD1-116 E	3.2262
936291	AD2-038 C O1	5.0747
936292	AD2-038 E O1	33.9613
938851	AE1-113 C O1	36.2967
938852	AE1-113 E O1	128.6883
939321	AE1-163 C O1	12.7518
939322	AE1-163 E O1	78.3322
942421	AE2-255 C O1	13.7488
942422	AE2-255 E O1	41.2463
942651	AE2-281 C O1	1.8217
942652	AE2-281 E O1	11.1903
964951	J974 C	6.2166
964952	J974 E	33.6334
965681	J1055 C	2.3140
965682	J1055 E	12.5194
LGEE	LGEE	0.1461
CIN	CIN	1.1319
CPLE	CPLE	0.1971
IPL	IPL	0.5733
G-007	G-007	0.0598
CBM-W2	CBM-W2	42.3261
CBM-W1	CBM-W1	3.4393
O-066	O-066	0.3903
CBM-S2	CBM-S2	0.6626
CARR	CARR	0.0272
CBM-S1	CBM-S1	2.8497
MEC	MEC	15.3245
Z1-043	Z1-043	0.0000

Bus #	Bus	MW Impact
CHOCTAW /* 35% REVERSE 4566958 4511400	CHOCTAW /* 35% REVERSE 4566958 4511400	0.0000
RENSSELAER	RENSSELAER	0.0212

11.6.2 Index 2

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
8899933	270716	DRESDEN ; B	CE	275179	DRESDEN ;1M	CE	1	COMED_P4_012-45-BT5-6__	breaker	442.0	130.74	133.54	AC	12.2

Bus #	Bus	MW Impact
274658	DRESDEN ;2U	27.1598
274879	MINONK ;1U	0.6464
290021	O50 E	19.5232
916512	Z1-107 E	1.4912
930751	AB1-122 2O1	128.5297
934111	AD1-039 2	12.5959
938851	AE1-113 C O1	8.0533
938852	AE1-113 E O1	28.5527
942421	AE2-255 C O1	3.0505
942422	AE2-255 E O1	9.1515
NEWTON	NEWTON	0.0249
CHILHOWEE	CHILHOWEE	0.0054
G-007	G-007	0.2027
CBM-W2	CBM-W2	3.3719
CBM-W1	CBM-W1	4.0240
WEC	WEC	1.0547
O-066	O-066	1.2989
CARR	CARR	0.0720
CHEOAH	CHEOAH	0.0165
CBM-S1	CBM-S1	0.0419
TILTON	TILTON	0.1619
SANTEETLA	SANTEETLA	0.0050
HAMLET	HAMLET	0.0586
BLUEG	BLUEG	0.8968
MEC	MEC	5.4054
CANNELTON	CANNELTON	0.0356
GIBSON	GIBSON	0.0311
CALDERWOOD	CALDERWOOD	0.0169
TRIMBLE	TRIMBLE	0.1024
RENSSELAER	RENSSELAER	0.0569
CATAWBA	CATAWBA	0.0319

11.6.3 Index 3

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
8900990	270716	DRESDEN ;B	CE	274702	KENDALL ;BU	CE	1	COMED_P7_345-L11620_B-S+_345-L11622_R-S	tower	1195.0	112.52	113.72	DC	16.69

Bus #	Bus	MW Impact
274658	DRESDEN ;2U	35.8168
274728	ELWOOD EC;5P	3.7633
274730	ELWOOD EC;6P	3.7633
274732	ELWOOD EC;7P	3.7633
274734	ELWOOD EC;8P	3.7633
274736	ELWOOD EC;9P	3.7633
290021	O50 E	22.6984
927091	AC1-204 1	105.6980
927101	AC1-204 2	89.2316
930741	AB1-122 101	39.9455
930751	AB1-122 201	169.6193
934101	AD1-039 1	3.9147
934111	AD1-039 2	16.6227
936511	AD2-066 C O1	7.8204
936512	AD2-066 E O1	5.2136
937401	AD2-194 1	11.3664
937411	AD2-194 2	9.5957
938511	AE1-070 1	13.3555
938521	AE1-070 2	10.3153
938851	AE1-113 C O1	9.3631
938852	AE1-113 E O1	33.1964
942421	AE2-255 C O1	3.5466
942422	AE2-255 E O1	10.6399
CIN	CIN	0.4777
IPL	IPL	0.1795
G-007	G-007	0.2310
CBM-W2	CBM-W2	20.2788
CBM-W1	CBM-W1	0.5443
O-066	O-066	1.4873
CBM-S2	CBM-S2	0.1017
CARR	CARR	0.0849
CBM-S1	CBM-S1	1.1399
BLUEG	BLUEG	0.1854
MEC	MEC	7.9880
TRIMBLE	TRIMBLE	0.0312
RENSSELAER	RENSSELAER	0.0671

11.6.4 Index 4

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
8900980	270855	POWERTON ; R	CE	349662	7TAZEWELL	AMIL	1	COMED_P7_345-L19601_B-S+_345-L9801_R-S_FSA	tower	1479.0	131.33	134.31	DC	44.12

Bus #	Bus	MW Impact
274677	POWERTON ;5U	74.2440
274678	POWERTON ;6U	74.4263
274861	TOP CROP ;1U	1.9698
274862	TOP CROP ;2U	3.8238
274879	MINONK ;1U	2.3369
290021	O50 E	70.5840
293644	O22 E1	59.4962
293645	O22 E2	115.4926
918051	AA1-018 C	1.8876
918052	AA1-018 E	95.3851
934871	AD1-116 C	5.5550
934872	AD1-116 E	9.0634
936291	AD2-038 C O1	14.2518
936292	AD2-038 E O1	95.3772
938851	AE1-113 C O1	29.1159
938852	AE1-113 E O1	103.2291
939321	AE1-163 C O1	35.8121
939322	AE1-163 E O1	219.9889
942421	AE2-255 C O1	11.0288
942422	AE2-255 E O1	33.0863
942651	AE2-281 C O1	5.1160
942652	AE2-281 E O1	31.4270
DUCKCREEK	DUCKCREEK	17.1666
NEWTON	NEWTON	3.3111
CHILHOWEE	CHILHOWEE	0.2811
G-007	G-007	0.8862
BAYOU /* 35% REVERSE 4392716	BAYOU /* 35% REVERSE 4392716	0.0000
BIG_CAJUN1 /* 35% REVERSE 4646887 4392735	BIG_CAJUN1 /* 35% REVERSE 4646887 4392735	0.0000
TVA	TVA	3.4212
PRAIRIE	PRAIRIE	8.3198
AE1-080A	AE1-080A	0.0000
WEC	WEC	0.2088
O-066	O-066	5.6734
COFFEEN	COFFEEN	1.7641
CARR	CARR	0.3039
EDWARDS	EDWARDS	8.6016
CHEOAH	CHEOAH	0.7823
TILTON	TILTON	1.2027
SANTEETLA	SANTEETLA	0.2289

Bus #	Bus	MW Impact
SMITHLAND	SMITHLAND	0.4733
COTTONWOOD	COTTONWOOD	5.1240
HAMLET	HAMLET	0.7164
BLUEG	BLUEG	5.4108
UNIONPOWER	UNIONPOWER	1.7702
CANNELTON	CANNELTON	0.4357
GIBSON	GIBSON	0.2967
CALDERWOOD	CALDERWOOD	0.8578
FARMERCITY	FARMERCITY	1.3440
TRIMBLE	TRIMBLE	0.5838
RENSSELAER	RENSSELAER	0.2405
CATAWBA	CATAWBA	0.4536

11.6.5 Index 5

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
8899930	270926	WILTON ; B	CE	275232	WILTON ;3M	CE	1	COMED_P4_112-65-BT5-6__	breaker	1379.0	148.48	148.72	AC	14.81

Bus #	Bus	MW Impact
270859	PWR VTR EC;R	12.5647
274722	S-055 E	11.6685
274772	LINCOLN ;3U	2.8996
274773	LINCOLN ;4U	2.8996
274774	LINCOLN ;5U	2.8996
274775	LINCOLN ;6U	2.8996
274776	LINCOLN ;7U	2.8996
274777	LINCOLN ;8U	2.8996
274859	EASYR;U1 E	11.4199
274860	EASYR;U2 E	11.4199
274888	PILOT HIL;1E	20.0408
274890	CAYUG;1U E	17.1870
274891	CAYUG;2U E	17.1870
275149	KEMPTON ;1E	20.0408
290021	O50 E	20.1470
290051	GSG-6; E	10.8607
290108	LEEDK;1U E	25.2331
293061	N-015 E	16.4995
293516	O-009 E1	9.4483
293517	O-009 E2	4.7990
293518	O-009 E3	5.2850
293644	O22 E1	10.6516
293645	O22 E2	20.6767
293715	O-029 E	10.1639
293716	O-029 E	5.5727
293717	O-029 E	5.1219
293771	O-035 E	6.7117
294392	P-010 E	20.9544
294763	P-046 E	9.7362
295109	WESTBROOK E	5.8147
295111	SUBLETTE E	2.6936
910542	X3-005 E	0.7597
914641	Y2-103 (Withdrawn : 12/24/2019)	46.6742
915011	Y3-013 1	3.8895
915021	Y3-013 2	3.8895
915031	Y3-013 3	3.8895
916211	Z1-072 E	5.0791
916221	Z1-073 E	5.6039
916502	Z1-106 E1	1.5440
916504	Z1-106 E2	1.5441
916512	Z1-107 E	3.1616
916522	Z1-108 E	3.0384

Bus #	Bus	MW Impact
917502	Z2-087 E	21.8537
918052	AA1-018 E	17.0791
919221	AA1-146	18.2733
919581	AA2-030	18.2733
920272	AA2-123 E	2.9858
924041	AB2-047 C O1	4.0819
924042	AB2-047 E O1	27.3171
924471	AB2-096	43.9764
925161	AB2-173 (Withdrawn : 12/24/2019)	3.2589
925302	AB2-191 E	1.4382
925581	AC1-033 C	1.4679
925582	AC1-033 E	9.8273
926311	AC1-109 1	1.9866
926321	AC1-109 2	1.9866
926331	AC1-110 1	1.9726
926341	AC1-110 2	1.9726
926351	AC1-111 1	0.7934
926361	AC1-111 2	0.7934
926371	AC1-111 3	0.7934
926381	AC1-111 4	0.7934
926391	AC1-111 5	0.7934
926401	AC1-111 6	0.7934
926431	AC1-114	2.4709
926821	AC1-168 C O1	1.2125
926822	AC1-168 E O1	8.1368
927091	AC1-204 1	75.4544
927101	AC1-204 2	75.4544
927201	AC1-214 C O1	2.1541
927202	AC1-214 E O1	6.8477
927451	AC1-142A 1	4.3471
927461	AC1-142A 2	4.3468
927511	AC1-113 1	1.2355
927521	AC1-113 2	1.2355
930481	AB1-089	68.2924
930501	AB1-091 O1	79.7415
930741	AB1-122 1O1	75.8149
930751	AB1-122 2O1	76.6262
932881	AC2-115 1	2.4709
932891	AC2-115 2	2.4709
932921	AC2-116	0.8648
932931	AC2-117	5.5346
933341	AC2-147 C	0.9063
933342	AC2-147 E	1.4788
933411	AC2-154 C	2.7198
933412	AC2-154 E	4.4376
933431	AC2-156 C O1	0.9941
933432	AC2-156 E O1	1.6219
933911	AD1-013 C	1.9121
933912	AD1-013 E	3.0543
933931	AD1-016 C	0.9644
933932	AD1-016 E	1.5735
934101	AD1-039 1	7.4299

Bus #	Bus	MW Impact
934111	AD1-039 2	7.5094
934431	AD1-067 C	0.1364
934432	AD1-067 E	0.5734
934651	AD1-096 C	0.9260
934652	AD1-096 E	1.5109
934701	AD1-098 C O1	7.1474
934702	AD1-098 E O1	5.2184
934721	AD1-100 C	24.9836
934722	AD1-100 E	116.5903
934871	AD1-116 C	0.9946
934872	AD1-116 E	1.6228
934971	AD1-129 C	0.9387
934972	AD1-129 E	0.6258
935001	AD1-133 C O1	23.2606
935002	AD1-133 E O1	15.5071
936291	AD2-038 C O1	2.4451
936292	AD2-038 E O1	16.3637
936371	AD2-047 C O1	4.8670
936372	AD2-047 E O1	23.7627
936461	AD2-060	2.8630
936511	AD2-066 C O1	8.7894
936512	AD2-066 E O1	5.8596
936791	AD2-102 C	14.6849
936792	AD2-102 E	9.7900
937001	AD2-134 C	2.8392
937002	AD2-134 E	11.7286
937311	AD2-172 C	2.5558
937312	AD2-172 E	3.5294
937321	AD2-175 C (Withdrawn : 12/10/2019)	17.8446
937322	AD2-175 E (Withdrawn : 12/10/2019)	11.8964
937401	AD2-194 1	8.1141
937411	AD2-194 2	8.1141
937531	AD2-214 C	4.5875
937532	AD2-214 E	3.0583
938511	AE1-070 1	9.5341
938521	AE1-070 2	8.7227
938851	AE1-113 C O1	8.3107
938852	AE1-113 E O1	29.4650
938861	AE1-114 C O1	4.0836
938862	AE1-114 E O1	13.9322
939051	AE1-134 1	1.4200
939061	AE1-134 2	1.4200
939321	AE1-163 C O1	6.1442
939322	AE1-163 E O1	37.7430
939351	AE1-166 C O1	12.3265
939352	AE1-166 E O1	11.3783
939401	AE1-172 C O1	6.8666
939402	AE1-172 E O1	32.1484
939631	AE1-193 C O1	0.1763
939632	AE1-193 E O1	1.1799
939681	AE1-198 C O1	21.6251
939682	AE1-198 E O1	18.3757

Bus #	Bus	MW Impact
939731	AE1-204 E (Withdrawn : 11/18/2019)	0.5724
939741	AE1-205 C O1	10.5843
939742	AE1-205 E O1	14.6165
940101	AE1-252 C O1	13.7700
940102	AE1-252 E O1	9.1800
940501	AE2-035 C	2.5558
940502	AE2-035 E	3.5294
940621	AE2-049 C O1	9.8022
940622	AE2-049 E O1	6.5348
940631	AE2-050 C O1	13.0784
940632	AE2-050 E O1	8.7190
940752	AE2-062 E	0.1357
940762	AE2-063 E (Withdrawn : 01/14/2020)	0.1357
941131	AE2-107 C	7.5949
941132	AE2-107 E	5.0633
941551	AE2-152 C O1	14.2229
941552	AE2-152 E O1	9.4819
941561	AE2-153 C O1	5.1510
941562	AE2-153 E O1	24.1162
941731	AE2-173 O1	6.2798
942111	AE2-223 C	2.4491
942112	AE2-223 E	16.3903
942421	AE2-255 C O1	3.1480
942422	AE2-255 E O1	9.4439
942651	AE2-281 C O1	0.8777
942652	AE2-281 E O1	5.3919
942881	AE2-307 C O1	23.9771
942882	AE2-307 E O1	8.7190
942911	AE2-310 C O1	9.5181
942912	AE2-310 E O1	2.5571
942991	AE2-321 C	8.4907
942992	AE2-321 E	4.1820
943121	AE2-341 C	13.2720
943122	AE2-341 E	6.5173
CHILHOWEE	CHILHOWEE	0.0343
G-007	G-007	2.7258
CBM-W2	CBM-W2	70.7400
CBM-W1	CBM-W1	37.1718
WEC	WEC	9.7283
O-066	O-066	17.4913
CARR	CARR	0.9702
CHEOAH	CHEOAH	0.1171
CBM-S1	CBM-S1	1.7291
SANTEETLA	SANTEETLA	0.0369
HAMLET	HAMLET	0.7429
BLUEG	BLUEG	8.0110
MEC	MEC	46.4402
CANNELTON	CANNELTON	0.0968
GIBSON	GIBSON	0.0075
Z1-043	Z1-043	0.0000
CALDERWOOD	CALDERWOOD	0.1125
TRIMBLE	TRIMBLE	0.9538

Bus #	Bus	MW Impact
CHOCTAW /* 35% REVERSE 4566958 4511400	CHOCTAW /* 35% REVERSE 4566958 4511400	0.0000
RENSSELAER	RENSSELAER	0.7669
CATAWBA	CATAWBA	0.3944

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ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
8899939	270927	WILTON ; R	CE	275233	WILTON ;4M	CE	1	COMED_P4_112-65-BT2-3__	breaker	1379.0	151.55	151.79	AC	15.13

Bus #	Bus	MW Impact
270859	PWR VTR EC;R	12.8316
274722	S-055 E	11.9179
274772	LINCOLN ;3U	2.9778
274773	LINCOLN ;4U	2.9778
274774	LINCOLN ;5U	2.9778
274775	LINCOLN ;6U	2.9778
274776	LINCOLN ;7U	2.9778
274777	LINCOLN ;8U	2.9778
274859	EASYR;U1 E	11.6631
274860	EASYR;U2 E	11.6631
274888	PILOT HIL;1E	20.4585
274890	CAYUG;1U E	17.5297
274891	CAYUG;2U E	17.5297
275149	KEMPTON ;1E	20.4585
290021	O50 E	20.5768
290051	GSG-6; E	11.0916
290108	LEEDK;1U E	25.7686
293061	N-015 E	16.8433
293516	O-009 E1	9.6484
293517	O-009 E2	4.9006
293518	O-009 E3	5.3969
293644	O22 E1	10.8770
293645	O22 E2	21.1142
293715	O-029 E	10.3793
293716	O-029 E	5.6908
293717	O-029 E	5.2305
293771	O-035 E	6.8531
294392	P-010 E	21.3909
294763	P-046 E	9.9430
295109	WESTBROOK E	5.9383
295111	SUBLETTE E	2.7508
910542	X3-005 E	0.7763
914641	Y2-103 (Withdrawn : 12/24/2019)	47.6717
915011	Y3-013 1	3.9726
915021	Y3-013 2	3.9726
915031	Y3-013 3	3.9726
916211	Z1-072 E	5.1861
916221	Z1-073 E	5.7230
916502	Z1-106 E1	1.5769
916504	Z1-106 E2	1.5770
916512	Z1-107 E	3.2290
916522	Z1-108 E	3.1032

Bus #	Bus	MW Impact
917502	Z2-087 E	22.2959
918052	AA1-018 E	17.4518
919221	AA1-146	18.6616
919581	AA2-030	18.6616
920272	AA2-123 E	3.0494
924041	AB2-047 C O1	4.1645
924042	AB2-047 E O1	27.8699
924471	AB2-096	44.9106
925161	AB2-173 (Withdrawn : 12/24/2019)	3.3282
925302	AB2-191 E	1.4687
925581	AC1-033 C	1.4989
925582	AC1-033 E	10.0345
926311	AC1-109 1	2.0289
926321	AC1-109 2	2.0289
926331	AC1-110 1	2.0145
926341	AC1-110 2	2.0145
926351	AC1-111 1	0.8103
926361	AC1-111 2	0.8103
926371	AC1-111 3	0.8103
926381	AC1-111 4	0.8103
926391	AC1-111 5	0.8103
926401	AC1-111 6	0.8103
926431	AC1-114	2.5235
926821	AC1-168 C O1	1.2380
926822	AC1-168 E O1	8.3079
927091	AC1-204 1	77.0716
927101	AC1-204 2	77.0668
927201	AC1-214 C O1	2.1995
927202	AC1-214 E O1	6.9920
927451	AC1-142A 1	4.4404
927461	AC1-142A 2	4.4404
927511	AC1-113 1	1.2617
927521	AC1-113 2	1.2617
930481	AB1-089	69.7416
930501	AB1-091 O1	81.3964
930741	AB1-122 1O1	77.4180
930751	AB1-122 2O1	78.2586
932881	AC2-115 1	2.5235
932891	AC2-115 2	2.5235
932921	AC2-116	0.8832
932931	AC2-117	5.6550
933341	AC2-147 C	0.9256
933342	AC2-147 E	1.5102
933411	AC2-154 C	2.7765
933412	AC2-154 E	4.5301
933431	AC2-156 C O1	1.0152
933432	AC2-156 E O1	1.6564
933911	AD1-013 C	1.9526
933912	AD1-013 E	3.1191
933931	AD1-016 C	0.9850
933932	AD1-016 E	1.6070
934101	AD1-039 1	7.5870

Bus #	Bus	MW Impact
934111	AD1-039 2	7.6693
934431	AD1-067 C	0.1393
934432	AD1-067 E	0.5856
934651	AD1-096 C	0.9457
934652	AD1-096 E	1.5431
934701	AD1-098 C O1	7.2992
934702	AD1-098 E O1	5.3292
934721	AD1-100 C	25.4783
934722	AD1-100 E	118.8988
934871	AD1-116 C	1.0164
934872	AD1-116 E	1.6583
934971	AD1-129 C	0.9587
934972	AD1-129 E	0.6391
935001	AD1-133 C O1	23.7441
935002	AD1-133 E O1	15.8294
936291	AD2-038 C O1	2.4979
936292	AD2-038 E O1	16.7164
936371	AD2-047 C O1	4.9685
936372	AD2-047 E O1	24.2579
936461	AD2-060	2.9226
936511	AD2-066 C O1	8.9752
936512	AD2-066 E O1	5.9834
936791	AD2-102 C	14.9971
936792	AD2-102 E	9.9980
937001	AD2-134 C	2.8995
937002	AD2-134 E	11.9779
937311	AD2-172 C	2.6100
937312	AD2-172 E	3.6043
937321	AD2-175 C (Withdrawn : 12/10/2019)	18.2147
937322	AD2-175 E (Withdrawn : 12/10/2019)	12.1431
937401	AD2-194 1	8.2880
937411	AD2-194 2	8.2875
937531	AD2-214 C	4.6849
937532	AD2-214 E	3.1233
938511	AE1-070 1	9.7384
938521	AE1-070 2	8.9091
938851	AE1-113 C O1	8.4879
938852	AE1-113 E O1	30.0936
938861	AE1-114 C O1	4.1706
938862	AE1-114 E O1	14.2290
939051	AE1-134 1	1.4501
939061	AE1-134 2	1.4501
939321	AE1-163 C O1	6.2767
939322	AE1-163 E O1	38.5566
939351	AE1-166 C O1	12.5745
939352	AE1-166 E O1	11.6072
939401	AE1-172 C O1	7.0024
939402	AE1-172 E O1	32.7842
939631	AE1-193 C O1	0.1800
939632	AE1-193 E O1	1.2048
939681	AE1-198 C O1	22.0811
939682	AE1-198 E O1	18.7632

Bus #	Bus	MW Impact
939731	AE1-204 E (Withdrawn : 11/18/2019)	0.5846
939741	AE1-205 C O1	10.7985
939742	AE1-205 E O1	14.9123
940101	AE1-252 C O1	14.0423
940102	AE1-252 E O1	9.3616
940501	AE2-035 C	2.6100
940502	AE2-035 E	3.6043
940621	AE2-049 C O1	10.0069
940622	AE2-049 E O1	6.6712
940631	AE2-050 C O1	13.3470
940632	AE2-050 E O1	8.8980
940752	AE2-062 E	0.1386
940762	AE2-063 E (Withdrawn : 01/14/2020)	0.1386
941131	AE2-107 C	7.7566
941132	AE2-107 E	5.1711
941551	AE2-152 C O1	14.5090
941552	AE2-152 E O1	9.6727
941561	AE2-153 C O1	5.2572
941562	AE2-153 E O1	24.6135
941731	AE2-173 O1	6.4069
942111	AE2-223 C	2.4987
942112	AE2-223 E	16.7219
942421	AE2-255 C O1	3.2151
942422	AE2-255 E O1	9.6454
942651	AE2-281 C O1	0.8967
942652	AE2-281 E O1	5.5081
942881	AE2-307 C O1	24.4694
942882	AE2-307 E O1	8.8980
942911	AE2-310 C O1	9.7168
942912	AE2-310 E O1	2.6105
942991	AE2-321 C	8.6712
942992	AE2-321 E	4.2709
943121	AE2-341 C	13.5533
943122	AE2-341 E	6.6554
CHILHOWEE	CHILHOWEE	0.0352
G-007	G-007	2.7846
CBM-W2	CBM-W2	72.2020
CBM-W1	CBM-W1	37.9650
WEC	WEC	9.9358
O-066	O-066	17.8614
CARR	CARR	0.9911
CHEOAH	CHEOAH	0.1201
CBM-S1	CBM-S1	1.7613
SANTEETLA	SANTEETLA	0.0378
HAMLET	HAMLET	0.7596
BLUEG	BLUEG	8.1863
MEC	MEC	47.4197
CANNELTON	CANNELTON	0.0993
GIBSON	GIBSON	0.0079
Z1-043	Z1-043	0.0000
CALDERWOOD	CALDERWOOD	0.1158
TRIMBLE	TRIMBLE	0.9750

Bus #	Bus	MW Impact
CHOCTAW /* 35% REVERSE 4566958 4511400	CHOCTAW /* 35% REVERSE 4566958 4511400	0.0000
RENSSELAER	RENSSELAER	0.7832
CATAWBA	CATAWBA	0.4032

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ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
8900111	274702	KENDALL ;BU	CE	270810	LOCKPORT ; B	CE	1	COMED_P4_111-45-L14321T	breaker	1479.0	107.33	108.5	DC	17.25

Bus #	Bus	MW Impact
274658	DRESDEN ;2U	37.6310
274704	KENDALL ;1C	11.9063
274705	KENDALL ;1S	7.9397
274706	KENDALL ;2C	11.9063
274707	KENDALL ;2S	7.9397
274836	EQUISTAR ; R	2.1625
274879	MINONK ;1U	0.9140
290021	O50 E	27.6048
904211	W3-135	0.1905
927091	AC1-204 1	45.9271
927101	AC1-204 2	45.2252
930751	AB1-122 2O1	178.1523
934111	AD1-039 2	17.4589
936511	AD2-066 C O1	9.0042
936512	AD2-066 E O1	6.0028
937401	AD2-194 1	4.9388
937411	AD2-194 2	4.8634
938511	AE1-070 1	5.8031
938521	AE1-070 2	5.2281
938851	AE1-113 C O1	11.3870
938852	AE1-113 E O1	40.3720
942421	AE2-255 C O1	4.3132
942422	AE2-255 E O1	12.9397
953201	J715 C	0.8600
953202	J715 E	4.6530
CHILHOWEE	CHILHOWEE	0.0293
G-007	G-007	0.3171
CBM-W2	CBM-W2	2.6881
TVA	TVA	0.1768
O-066	O-066	2.0325
CARR	CARR	0.1125
CHEOAH	CHEOAH	0.0836
SANTEETLA	SANTEETLA	0.0247
COTTONWOOD	COTTONWOOD	0.2121
HAMLET	HAMLET	0.1305
BLUEG	BLUEG	1.0020
UNIONPOWER	UNIONPOWER	0.0890
CANNELTON	CANNELTON	0.0341
GIBSON	GIBSON	0.0130
CALDERWOOD	CALDERWOOD	0.0901
FARMERCITY	FARMERCITY	0.1544
TRIMBLE	TRIMBLE	0.1158

Bus #	Bus	MW Impact
RENSSELAER	RENSSELAER	0.0889
CATAWBA	CATAWBA	0.0763

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ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
7290249	274804	UNIV PK N;RP	CE	243229	05OLIVE	AEP	1	AEP_P4_#2978_05DUMONT 765_B	breaker	971.0	130.82	131.36	AC	9.75

Bus #	Bus	MW Impact
270859	PWR VTR EC;R	8.3113
274722	S-055 E	7.7640
274808	UNIV PK N;4U	1.3951
274809	UNIV PK N;5U	1.3951
274811	UNIV PK N;7U	1.3951
274812	UNIV PK N;8U	1.3951
274814	UNIV PK N;0U	1.3951
274815	UNIV PK N;XU	1.3951
274859	EASYR;U1 E	7.5064
274860	EASYR;U2 E	7.5064
274888	PILOT HIL;1E	12.6116
274890	CAYUG;1U E	9.0770
274891	CAYUG;2U E	9.0770
275149	KEMPTON ;1E	12.6116
290021	O50 E	13.2559
290051	GSG-6; E	7.1534
290108	LEEDK;1U E	16.6413
293061	N-015 E	10.6967
293516	O-009 E1	6.1582
293517	O-009 E2	3.1279
293518	O-009 E3	3.4447
293644	O22 E1	6.6877
293645	O22 E2	12.9820
293715	O-029 E	6.6517
293716	O-029 E	3.6470
293717	O-029 E	3.3520
293771	O-035 E	4.3200
294392	P-010 E	13.5849
294401	BSHIL;1U E	5.7428
294410	BSHIL;2U E	5.7428
294763	P-046 E	6.4165
295109	WESTBROOK E	3.8298
295111	SUBLETTE E	1.7673
910542	X3-005 E	0.4357
914641	Y2-103 (Withdrawn : 12/24/2019)	31.0559
915011	Y3-013 1	2.5880
915021	Y3-013 2	2.5880
915031	Y3-013 3	2.5880
916211	Z1-072 E	3.2692
916221	Z1-073 E	3.6910
916502	Z1-106 E1	1.0227

Bus #	Bus	MW Impact
916504	Z1-106 E2	1.0228
916512	Z1-107 E	2.1702
916522	Z1-108 E	2.0152
918052	AA1-018 E	12.1341
919221	AA1-146	11.9811
919581	AA2-030	11.9811
919621	AA2-039 C	1.4072
919622	AA2-039 E	9.4175
920272	AA2-123 E	1.9766
924471	AB2-096	29.0895
925161	AB2-173 (Withdrawn : 12/24/2019)	2.1368
925302	AB2-191 E	0.9473
925581	AC1-033 C	0.9455
925582	AC1-033 E	6.3296
926311	AC1-109 1	1.3163
926321	AC1-109 2	1.3163
926331	AC1-110 1	1.3052
926341	AC1-110 2	1.3052
926351	AC1-111 1	0.5261
926361	AC1-111 2	0.5261
926371	AC1-111 3	0.5261
926381	AC1-111 4	0.5261
926391	AC1-111 5	0.5261
926401	AC1-111 6	0.5261
926431	AC1-114	1.6315
926821	AC1-168 C O1	0.7815
926822	AC1-168 E O1	5.2446
927091	AC1-204 1	50.2239
927101	AC1-204 2	50.2618
927201	AC1-214 C O1	1.3865
927202	AC1-214 E O1	4.4075
927451	AC1-142A 1	2.9308
927461	AC1-142A 2	2.9294
927511	AC1-113 1	0.8157
927521	AC1-113 2	0.8157
930481	AB1-089	45.0810
930501	AB1-091 O1	48.4844
930741	AB1-122 1O1	49.4077
930751	AB1-122 2O1	50.6589
932881	AC2-115 1	1.6315
932891	AC2-115 2	1.6315
932921	AC2-116	0.5710
932931	AC2-117	8.8264
933341	AC2-147 C	0.5956
933342	AC2-147 E	0.9718
933411	AC2-154 C	1.7117
933412	AC2-154 E	2.7928
933431	AC2-156 C O1	0.6566
933432	AC2-156 E O1	1.0713
933911	AD1-013 C	1.2595
933912	AD1-013 E	2.0120
933931	AD1-016 C	0.6384

Bus #	Bus	MW Impact
933932	AD1-016 E	1.0417
934051	AD1-031 C O1	1.9198
934052	AD1-031 E O1	3.1323
934101	AD1-039 1	4.8420
934111	AD1-039 2	4.9646
934431	AD1-067 C	0.0898
934432	AD1-067 E	0.3777
934651	AD1-096 C	0.6101
934652	AD1-096 E	0.9955
934701	AD1-098 C O1	4.7067
934702	AD1-098 E O1	3.4363
934721	AD1-100 C	13.2230
934722	AD1-100 E	61.7075
934871	AD1-116 C	0.7067
934872	AD1-116 E	1.1530
934971	AD1-129 C	0.6205
934972	AD1-129 E	0.4137
935001	AD1-133 C O1	14.1464
935002	AD1-133 E O1	9.4309
936291	AD2-038 C O1	1.6726
936292	AD2-038 E O1	11.1934
936371	AD2-047 C O1	3.0631
936372	AD2-047 E O1	14.9552
936461	AD2-060	1.8018
936511	AD2-066 C O1	5.7935
936512	AD2-066 E O1	3.8624
936791	AD2-102 C	9.6931
936792	AD2-102 E	6.4620
937001	AD2-134 C	1.8700
937002	AD2-134 E	7.7250
937311	AD2-172 C	1.6843
937312	AD2-172 E	2.3260
937321	AD2-175 C (Withdrawn : 12/10/2019)	10.8408
937322	AD2-175 E (Withdrawn : 12/10/2019)	7.2272
937401	AD2-194 1	5.4009
937411	AD2-194 2	5.4050
937531	AD2-214 C	2.9981
937532	AD2-214 E	1.9987
938511	AE1-070 1	6.3461
938521	AE1-070 2	5.8104
938851	AE1-113 C O1	5.4681
938852	AE1-113 E O1	19.3868
938861	AE1-114 C O1	2.6863
938862	AE1-114 E O1	9.1649
939051	AE1-134 1	0.9310
939061	AE1-134 2	0.9310
939321	AE1-163 C O1	4.2029
939322	AE1-163 E O1	25.8178
939351	AE1-166 C O1	7.0285
939352	AE1-166 E O1	6.4878
939401	AE1-172 C O1	3.5401
939402	AE1-172 E O1	16.5743

Bus #	Bus	MW Impact
939631	AE1-193 C O1	4.6913
939632	AE1-193 E O1	31.3955
939681	AE1-198 C O1	13.9295
939682	AE1-198 E O1	11.8365
939731	AE1-204 E (Withdrawn : 11/18/2019)	0.3809
940101	AE1-252 C O1	7.0992
940102	AE1-252 E O1	4.7328
940501	AE2-035 C	1.6843
940502	AE2-035 E	2.3260
940621	AE2-049 C O1	6.2122
940622	AE2-049 E O1	4.1415
940631	AE2-050 C O1	8.3194
940632	AE2-050 E O1	5.5463
940752	AE2-062 E	0.0883
940762	AE2-063 E (Withdrawn : 01/14/2020)	0.0883
941131	AE2-107 C	5.0102
941132	AE2-107 E	3.3402
941551	AE2-152 C O1	8.1098
941552	AE2-152 E O1	5.4065
941561	AE2-153 C O1	3.2481
941562	AE2-153 E O1	15.2071
942421	AE2-255 C O1	2.0712
942422	AE2-255 E O1	6.2137
942651	AE2-281 C O1	0.6004
942652	AE2-281 E O1	3.6883
942881	AE2-307 C O1	15.2522
942882	AE2-307 E O1	5.5463
942911	AE2-310 C O1	6.0321
942912	AE2-310 E O1	1.6206
942991	AE2-321 C	5.6198
942992	AE2-321 E	2.7680
943121	AE2-341 C	8.7682
943122	AE2-341 E	4.3057
CHILHOWEE	CHILHOWEE	0.0166
G-007	G-007	1.7314
CBM-W2	CBM-W2	43.0335
CBM-W1	CBM-W1	21.8341
WEC	WEC	6.4525
O-066	O-066	11.1112
CARR	CARR	0.6207
CHEOAH	CHEOAH	0.0596
CBM-S1	CBM-S1	1.1560
SANTEETLA	SANTEETLA	0.0190
HAMLET	HAMLET	0.4479
BLUEG	BLUEG	4.4890
MEC	MEC	30.0069
CANNELTON	CANNELTON	0.0667
GIBSON	GIBSON	0.0277
Z1-043	Z1-043	0.0000
CALDERWOOD	CALDERWOOD	0.0557
TRIMBLE	TRIMBLE	0.5303
CHOCTAW REVERSE 4566958 4511400	CHOCTAW REVERSE 4566958 4511400	0.0000

Bus #	Bus	MW Impact
RENSSELAER	RENSSELAER	0.4907
CATAWBA	CATAWBA	0.2352

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ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
8899941	275179	DRESDEN ;1M	CE	271337	DRESDEN ; R	CE	1	COMED_P4_012-45-BT5-6__	breaker	442.0	128.05	130.76	AC	12.2

Bus #	Bus	MW Impact
274658	DRESDEN ;2U	27.1598
274879	MINONK ;1U	0.6464
290021	O50 E	19.5232
916512	Z1-107 E	1.4912
930751	AB1-122 2O1	128.5297
934111	AD1-039 2	12.5959
938851	AE1-113 C O1	8.0533
938852	AE1-113 E O1	28.5527
942421	AE2-255 C O1	3.0505
942422	AE2-255 E O1	9.1515
NEWTON	NEWTON	0.0249
CHILHOWEE	CHILHOWEE	0.0054
G-007	G-007	0.2027
CBM-W2	CBM-W2	3.3719
CBM-W1	CBM-W1	4.0240
WEC	WEC	1.0547
O-066	O-066	1.2989
CARR	CARR	0.0720
CHEOAH	CHEOAH	0.0165
CBM-S1	CBM-S1	0.0419
TILTON	TILTON	0.1619
SANTEETLA	SANTEETLA	0.0050
HAMLET	HAMLET	0.0586
BLUEG	BLUEG	0.8968
MEC	MEC	5.4054
CANNELTON	CANNELTON	0.0356
GIBSON	GIBSON	0.0311
CALDERWOOD	CALDERWOOD	0.0169
TRIMBLE	TRIMBLE	0.1024
RENSSELAER	RENSSELAER	0.0569
CATAWBA	CATAWBA	0.0319

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ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
8899932	275232	WILTON ;3M	CE	270644	WILTON ;	CE	1	COMED_P4_112-65-BT5-6__	breaker	1379.0	148.37	148.61	AC	14.81

Bus #	Bus	MW Impact
270859	PWR VTR EC;R	12.5647
274722	S-055 E	11.6685
274772	LINCOLN ;3U	2.8996
274773	LINCOLN ;4U	2.8996
274774	LINCOLN ;5U	2.8996
274775	LINCOLN ;6U	2.8996
274776	LINCOLN ;7U	2.8996
274777	LINCOLN ;8U	2.8996
274859	EASYR;U1 E	11.4199
274860	EASYR;U2 E	11.4199
274888	PILOT HIL;1E	20.0408
274890	CAYUG;1U E	17.1870
274891	CAYUG;2U E	17.1870
275149	KEMPTON ;1E	20.0408
290021	O50 E	20.1470
290051	GSG-6; E	10.8607
290108	LEEDK;1U E	25.2331
293061	N-015 E	16.4995
293516	O-009 E1	9.4483
293517	O-009 E2	4.7990
293518	O-009 E3	5.2850
293644	O22 E1	10.6516
293645	O22 E2	20.6767
293715	O-029 E	10.1639
293716	O-029 E	5.5727
293717	O-029 E	5.1219
293771	O-035 E	6.7117
294392	P-010 E	20.9544
294763	P-046 E	9.7362
295109	WESTBROOK E	5.8147
295111	SUBLETTE E	2.6936
910542	X3-005 E	0.7597
914641	Y2-103 (Withdrawn : 12/24/2019)	46.6742
915011	Y3-013 1	3.8895
915021	Y3-013 2	3.8895
915031	Y3-013 3	3.8895
916211	Z1-072 E	5.0791
916221	Z1-073 E	5.6039
916502	Z1-106 E1	1.5440
916504	Z1-106 E2	1.5441
916512	Z1-107 E	3.1616
916522	Z1-108 E	3.0384

Bus #	Bus	MW Impact
917502	Z2-087 E	21.8537
918052	AA1-018 E	17.0791
919221	AA1-146	18.2733
919581	AA2-030	18.2733
920272	AA2-123 E	2.9858
924041	AB2-047 C O1	4.0819
924042	AB2-047 E O1	27.3171
924471	AB2-096	43.9764
925161	AB2-173 (Withdrawn : 12/24/2019)	3.2589
925302	AB2-191 E	1.4382
925581	AC1-033 C	1.4679
925582	AC1-033 E	9.8273
926311	AC1-109 1	1.9866
926321	AC1-109 2	1.9866
926331	AC1-110 1	1.9726
926341	AC1-110 2	1.9726
926351	AC1-111 1	0.7934
926361	AC1-111 2	0.7934
926371	AC1-111 3	0.7934
926381	AC1-111 4	0.7934
926391	AC1-111 5	0.7934
926401	AC1-111 6	0.7934
926431	AC1-114	2.4709
926821	AC1-168 C O1	1.2125
926822	AC1-168 E O1	8.1368
927091	AC1-204 1	75.4544
927101	AC1-204 2	75.4544
927201	AC1-214 C O1	2.1541
927202	AC1-214 E O1	6.8477
927451	AC1-142A 1	4.3471
927461	AC1-142A 2	4.3468
927511	AC1-113 1	1.2355
927521	AC1-113 2	1.2355
930481	AB1-089	68.2924
930501	AB1-091 O1	79.7415
930741	AB1-122 1O1	75.8149
930751	AB1-122 2O1	76.6262
932881	AC2-115 1	2.4709
932891	AC2-115 2	2.4709
932921	AC2-116	0.8648
932931	AC2-117	5.5346
933341	AC2-147 C	0.9063
933342	AC2-147 E	1.4788
933411	AC2-154 C	2.7198
933412	AC2-154 E	4.4376
933431	AC2-156 C O1	0.9941
933432	AC2-156 E O1	1.6219
933911	AD1-013 C	1.9121
933912	AD1-013 E	3.0543
933931	AD1-016 C	0.9644
933932	AD1-016 E	1.5735
934101	AD1-039 1	7.4299

Bus #	Bus	MW Impact
934111	AD1-039 2	7.5094
934431	AD1-067 C	0.1364
934432	AD1-067 E	0.5734
934651	AD1-096 C	0.9260
934652	AD1-096 E	1.5109
934701	AD1-098 C O1	7.1474
934702	AD1-098 E O1	5.2184
934721	AD1-100 C	24.9836
934722	AD1-100 E	116.5903
934871	AD1-116 C	0.9946
934872	AD1-116 E	1.6228
934971	AD1-129 C	0.9387
934972	AD1-129 E	0.6258
935001	AD1-133 C O1	23.2606
935002	AD1-133 E O1	15.5071
936291	AD2-038 C O1	2.4451
936292	AD2-038 E O1	16.3637
936371	AD2-047 C O1	4.8670
936372	AD2-047 E O1	23.7627
936461	AD2-060	2.8630
936511	AD2-066 C O1	8.7894
936512	AD2-066 E O1	5.8596
936791	AD2-102 C	14.6849
936792	AD2-102 E	9.7900
937001	AD2-134 C	2.8392
937002	AD2-134 E	11.7286
937311	AD2-172 C	2.5558
937312	AD2-172 E	3.5294
937321	AD2-175 C (Withdrawn : 12/10/2019)	17.8446
937322	AD2-175 E (Withdrawn : 12/10/2019)	11.8964
937401	AD2-194 1	8.1141
937411	AD2-194 2	8.1141
937531	AD2-214 C	4.5875
937532	AD2-214 E	3.0583
938511	AE1-070 1	9.5341
938521	AE1-070 2	8.7227
938851	AE1-113 C O1	8.3107
938852	AE1-113 E O1	29.4650
938861	AE1-114 C O1	4.0836
938862	AE1-114 E O1	13.9322
939051	AE1-134 1	1.4200
939061	AE1-134 2	1.4200
939321	AE1-163 C O1	6.1442
939322	AE1-163 E O1	37.7430
939351	AE1-166 C O1	12.3265
939352	AE1-166 E O1	11.3783
939401	AE1-172 C O1	6.8666
939402	AE1-172 E O1	32.1484
939631	AE1-193 C O1	0.1763
939632	AE1-193 E O1	1.1799
939681	AE1-198 C O1	21.6251
939682	AE1-198 E O1	18.3757

Bus #	Bus	MW Impact
939731	AE1-204 E (Withdrawn : 11/18/2019)	0.5724
939741	AE1-205 C O1	10.5843
939742	AE1-205 E O1	14.6165
940101	AE1-252 C O1	13.7700
940102	AE1-252 E O1	9.1800
940501	AE2-035 C	2.5558
940502	AE2-035 E	3.5294
940621	AE2-049 C O1	9.8022
940622	AE2-049 E O1	6.5348
940631	AE2-050 C O1	13.0784
940632	AE2-050 E O1	8.7190
940752	AE2-062 E	0.1357
940762	AE2-063 E (Withdrawn : 01/14/2020)	0.1357
941131	AE2-107 C	7.5949
941132	AE2-107 E	5.0633
941551	AE2-152 C O1	14.2229
941552	AE2-152 E O1	9.4819
941561	AE2-153 C O1	5.1510
941562	AE2-153 E O1	24.1162
941731	AE2-173 O1	6.2798
942111	AE2-223 C	2.4491
942112	AE2-223 E	16.3903
942421	AE2-255 C O1	3.1480
942422	AE2-255 E O1	9.4439
942651	AE2-281 C O1	0.8777
942652	AE2-281 E O1	5.3919
942881	AE2-307 C O1	23.9771
942882	AE2-307 E O1	8.7190
942911	AE2-310 C O1	9.5181
942912	AE2-310 E O1	2.5571
942991	AE2-321 C	8.4907
942992	AE2-321 E	4.1820
943121	AE2-341 C	13.2720
943122	AE2-341 E	6.5173
CHILHOWEE	CHILHOWEE	0.0343
G-007	G-007	2.7258
CBM-W2	CBM-W2	70.7400
CBM-W1	CBM-W1	37.1718
WEC	WEC	9.7283
O-066	O-066	17.4913
CARR	CARR	0.9702
CHEOAH	CHEOAH	0.1171
CBM-S1	CBM-S1	1.7291
SANTEETLA	SANTEETLA	0.0369
HAMLET	HAMLET	0.7429
BLUEG	BLUEG	8.0110
MEC	MEC	46.4402
CANNELTON	CANNELTON	0.0968
GIBSON	GIBSON	0.0075
Z1-043	Z1-043	0.0000
CALDERWOOD	CALDERWOOD	0.1125
TRIMBLE	TRIMBLE	0.9538

Bus #	Bus	MW Impact
CHOCTAW /* 35% REVERSE 4566958 4511400	CHOCTAW /* 35% REVERSE 4566958 4511400	0.0000
RENSSELAER	RENSSELAER	0.7669
CATAWBA	CATAWBA	0.3944

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ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
8899938	275233	WILTON ;4M	CE	270644	WILTON ;	CE	1	COMED_P4_112-65-BT2-3__	breaker	1379.0	151.43	151.67	AC	15.13

Bus #	Bus	MW Impact
270859	PWR VTR EC;R	12.8316
274722	S-055 E	11.9179
274772	LINCOLN ;3U	2.9778
274773	LINCOLN ;4U	2.9778
274774	LINCOLN ;5U	2.9778
274775	LINCOLN ;6U	2.9778
274776	LINCOLN ;7U	2.9778
274777	LINCOLN ;8U	2.9778
274859	EASYR;U1 E	11.6631
274860	EASYR;U2 E	11.6631
274888	PILOT HIL;1E	20.4585
274890	CAYUG;1U E	17.5297
274891	CAYUG;2U E	17.5297
275149	KEMPTON ;1E	20.4585
290021	O50 E	20.5768
290051	GSG-6; E	11.0916
290108	LEEDK;1U E	25.7686
293061	N-015 E	16.8433
293516	O-009 E1	9.6484
293517	O-009 E2	4.9006
293518	O-009 E3	5.3969
293644	O22 E1	10.8770
293645	O22 E2	21.1142
293715	O-029 E	10.3793
293716	O-029 E	5.6908
293717	O-029 E	5.2305
293771	O-035 E	6.8531
294392	P-010 E	21.3909
294763	P-046 E	9.9430
295109	WESTBROOK E	5.9383
295111	SUBLETTE E	2.7508
910542	X3-005 E	0.7763
914641	Y2-103 (Withdrawn : 12/24/2019)	47.6717
915011	Y3-013 1	3.9726
915021	Y3-013 2	3.9726
915031	Y3-013 3	3.9726
916211	Z1-072 E	5.1861
916221	Z1-073 E	5.7230
916502	Z1-106 E1	1.5769
916504	Z1-106 E2	1.5770
916512	Z1-107 E	3.2290
916522	Z1-108 E	3.1032

Bus #	Bus	MW Impact
917502	Z2-087 E	22.2959
918052	AA1-018 E	17.4518
919221	AA1-146	18.6616
919581	AA2-030	18.6616
920272	AA2-123 E	3.0494
924041	AB2-047 C O1	4.1645
924042	AB2-047 E O1	27.8699
924471	AB2-096	44.9106
925161	AB2-173 (Withdrawn : 12/24/2019)	3.3282
925302	AB2-191 E	1.4687
925581	AC1-033 C	1.4989
925582	AC1-033 E	10.0345
926311	AC1-109 1	2.0289
926321	AC1-109 2	2.0289
926331	AC1-110 1	2.0145
926341	AC1-110 2	2.0145
926351	AC1-111 1	0.8103
926361	AC1-111 2	0.8103
926371	AC1-111 3	0.8103
926381	AC1-111 4	0.8103
926391	AC1-111 5	0.8103
926401	AC1-111 6	0.8103
926431	AC1-114	2.5235
926821	AC1-168 C O1	1.2380
926822	AC1-168 E O1	8.3079
927091	AC1-204 1	77.0716
927101	AC1-204 2	77.0668
927201	AC1-214 C O1	2.1995
927202	AC1-214 E O1	6.9920
927451	AC1-142A 1	4.4404
927461	AC1-142A 2	4.4404
927511	AC1-113 1	1.2617
927521	AC1-113 2	1.2617
930481	AB1-089	69.7416
930501	AB1-091 O1	81.3964
930741	AB1-122 1O1	77.4180
930751	AB1-122 2O1	78.2586
932881	AC2-115 1	2.5235
932891	AC2-115 2	2.5235
932921	AC2-116	0.8832
932931	AC2-117	5.6550
933341	AC2-147 C	0.9256
933342	AC2-147 E	1.5102
933411	AC2-154 C	2.7765
933412	AC2-154 E	4.5301
933431	AC2-156 C O1	1.0152
933432	AC2-156 E O1	1.6564
933911	AD1-013 C	1.9526
933912	AD1-013 E	3.1191
933931	AD1-016 C	0.9850
933932	AD1-016 E	1.6070
934101	AD1-039 1	7.5870

Bus #	Bus	MW Impact
934111	AD1-039 2	7.6693
934431	AD1-067 C	0.1393
934432	AD1-067 E	0.5856
934651	AD1-096 C	0.9457
934652	AD1-096 E	1.5431
934701	AD1-098 C O1	7.2992
934702	AD1-098 E O1	5.3292
934721	AD1-100 C	25.4783
934722	AD1-100 E	118.8988
934871	AD1-116 C	1.0164
934872	AD1-116 E	1.6583
934971	AD1-129 C	0.9587
934972	AD1-129 E	0.6391
935001	AD1-133 C O1	23.7441
935002	AD1-133 E O1	15.8294
936291	AD2-038 C O1	2.4979
936292	AD2-038 E O1	16.7164
936371	AD2-047 C O1	4.9685
936372	AD2-047 E O1	24.2579
936461	AD2-060	2.9226
936511	AD2-066 C O1	8.9752
936512	AD2-066 E O1	5.9834
936791	AD2-102 C	14.9971
936792	AD2-102 E	9.9980
937001	AD2-134 C	2.8995
937002	AD2-134 E	11.9779
937311	AD2-172 C	2.6100
937312	AD2-172 E	3.6043
937321	AD2-175 C (Withdrawn : 12/10/2019)	18.2147
937322	AD2-175 E (Withdrawn : 12/10/2019)	12.1431
937401	AD2-194 1	8.2880
937411	AD2-194 2	8.2875
937531	AD2-214 C	4.6849
937532	AD2-214 E	3.1233
938511	AE1-070 1	9.7384
938521	AE1-070 2	8.9091
938851	AE1-113 C O1	8.4879
938852	AE1-113 E O1	30.0936
938861	AE1-114 C O1	4.1706
938862	AE1-114 E O1	14.2290
939051	AE1-134 1	1.4501
939061	AE1-134 2	1.4501
939321	AE1-163 C O1	6.2767
939322	AE1-163 E O1	38.5566
939351	AE1-166 C O1	12.5745
939352	AE1-166 E O1	11.6072
939401	AE1-172 C O1	7.0024
939402	AE1-172 E O1	32.7842
939631	AE1-193 C O1	0.1800
939632	AE1-193 E O1	1.2048
939681	AE1-198 C O1	22.0811
939682	AE1-198 E O1	18.7632

Bus #	Bus	MW Impact
939731	AE1-204 E (Withdrawn : 11/18/2019)	0.5846
939741	AE1-205 C O1	10.7985
939742	AE1-205 E O1	14.9123
940101	AE1-252 C O1	14.0423
940102	AE1-252 E O1	9.3616
940501	AE2-035 C	2.6100
940502	AE2-035 E	3.6043
940621	AE2-049 C O1	10.0069
940622	AE2-049 E O1	6.6712
940631	AE2-050 C O1	13.3470
940632	AE2-050 E O1	8.8980
940752	AE2-062 E	0.1386
940762	AE2-063 E (Withdrawn : 01/14/2020)	0.1386
941131	AE2-107 C	7.7566
941132	AE2-107 E	5.1711
941551	AE2-152 C O1	14.5090
941552	AE2-152 E O1	9.6727
941561	AE2-153 C O1	5.2572
941562	AE2-153 E O1	24.6135
941731	AE2-173 O1	6.4069
942111	AE2-223 C	2.4987
942112	AE2-223 E	16.7219
942421	AE2-255 C O1	3.2151
942422	AE2-255 E O1	9.6454
942651	AE2-281 C O1	0.8967
942652	AE2-281 E O1	5.5081
942881	AE2-307 C O1	24.4694
942882	AE2-307 E O1	8.8980
942911	AE2-310 C O1	9.7168
942912	AE2-310 E O1	2.6105
942991	AE2-321 C	8.6712
942992	AE2-321 E	4.2709
943121	AE2-341 C	13.5533
943122	AE2-341 E	6.6554
CHILHOWEE	CHILHOWEE	0.0352
G-007	G-007	2.7846
CBM-W2	CBM-W2	72.2020
CBM-W1	CBM-W1	37.9650
WEC	WEC	9.9358
O-066	O-066	17.8614
CARR	CARR	0.9911
CHEOAH	CHEOAH	0.1201
CBM-S1	CBM-S1	1.7613
SANTEETLA	SANTEETLA	0.0378
HAMLET	HAMLET	0.7596
BLUEG	BLUEG	8.1863
MEC	MEC	47.4197
CANNELTON	CANNELTON	0.0993
GIBSON	GIBSON	0.0079
Z1-043	Z1-043	0.0000
CALDERWOOD	CALDERWOOD	0.1158
TRIMBLE	TRIMBLE	0.9750

Bus #	Bus	MW Impact
CHOCTAW /* 35% REVERSE 4566958 4511400	CHOCTAW /* 35% REVERSE 4566958 4511400	0.0000
RENSSELAER	RENSSELAER	0.7832
CATAWBA	CATAWBA	0.4032

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ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
8900942	930740	AB1-122 TAP	CE	270717	DRESDEN ; R	CE	1	COMED_P7_345-L19601_B-S_+345-L9801_R-S_FSA	tower	1195.0	133.98	134.88	AC	12.15

Bus #	Bus	MW Impact
274677	POWERTON ;5U	22.6640
274678	POWERTON ;6U	22.7281
274861	TOP CROP ;1U	0.6008
274862	TOP CROP ;2U	1.1663
274879	MINONK ;1U	0.6435
290021	O50 E	19.4368
293644	O22 E1	18.1470
293645	O22 E2	35.2266
293771	O-035 E	3.5536
294401	BSHIL;1U E	5.2802
294410	BSHIL;2U E	5.2802
916211	Z1-072 E	2.6892
918051	AA1-018 C	0.5759
918052	AA1-018 E	29.1041
919621	AA2-039 C	1.2870
919622	AA2-039 E	8.6133
925581	AC1-033 C	0.8651
925582	AC1-033 E	5.7914
926841	AC1-171 C O1	0.7203
926842	AC1-171 E O1	4.8111
927201	AC1-214 C O1	1.1405
927202	AC1-214 E O1	3.6256
930741	AB1-122 1O1	524.8197
934051	AD1-031 C O1	1.7543
934052	AD1-031 E O1	2.8623
934101	AD1-039 1	51.4323
934871	AD1-116 C	1.6950
934872	AD1-116 E	2.7654
936291	AD2-038 C O1	4.3505
936292	AD2-038 E O1	29.1146
938851	AE1-113 C O1	8.0177
938852	AE1-113 E O1	28.4263
939321	AE1-163 C O1	10.9319
939322	AE1-163 E O1	67.1531
939631	AE1-193 C O1	4.2868
939632	AE1-193 E O1	28.6889
939681	AE1-198 C O1	12.7286
939682	AE1-198 E O1	10.8160
942421	AE2-255 C O1	3.0370
942422	AE2-255 E O1	9.1110
942651	AE2-281 C O1	1.5617

Bus #	Bus	MW Impact
942652	AE2-281 E O1	9.5933
951631	J456 C	1.7410
951632	J456 E	9.4190
953951	J859	10.6682
954702	J844 E	27.3090
964951	J974 C	7.4186
964952	J974 E	40.1364
965681	J1055 C	2.6827
965682	J1055 E	14.5138
990901	L-005 E	8.0172
LGEE	LGEE	0.1229
CIN	CIN	1.0020
CPLE	CPLE	0.2117
IPL	IPL	0.4930
G-007	G-007	0.1008
CBM-W2	CBM-W2	48.8342
CBM-W1	CBM-W1	6.5774
O-066	O-066	0.6528
CBM-S2	CBM-S2	0.7331
CARR	CARR	0.0423
CBM-S1	CBM-S1	3.2425
MEC	MEC	20.1986
Z1-043	Z1-043	0.0000
CHOCTAW REVERSE 4566958 4511400	CHOCTAW REVERSE 4566958 4511400	0.0000
RENSSELAER	RENSSELAER	0.0334

11.7 Contingency Definitions

Contingency Name	Contingency Definition
COMED_P4_111-45-L14321T	CONTINGENCY 'COMED_P4_111-45-L14321T' TRIP BRANCH FROM BUS 270928 TO BUS 270730 CKT 1 / WOLFS ; B 345 ELEC JUNC; B 345 TRIP BRANCH FROM BUS 270928 TO BUS 272794 TO BUS 275334 CKT 1 / WOLFS ; B 345 WOLFS ; B 138 WOLFS ;1C 34.5 TRIP BRANCH FROM BUS 270730 TO BUS 270846 CKT 1 / ELEC JUNC; B 345 PLANO ; B 345 TRIP BRANCH FROM BUS 270730 TO BUS 270916 CKT 1 / ELEC JUNC; B 345 WAYNE ; B 345 DISCONNECT BUS 275239 / ELEC JUNC;2M 138 END
COMED_P4_111-45-L11126_	CONTINGENCY 'COMED_P4_111-45-L11126_' TRIP BRANCH FROM BUS 270730 TO BUS 270916 CKT 1 / ELEC JUNC; B 345 WAYNE ; B 345 TRIP BRANCH FROM BUS 270916 TO BUS 270917 CKT 1 / WAYNE ; B 345 WAYNE ; R 345 TRIP BRANCH FROM BUS 270730 TO BUS 270846 CKT 1 / ELEC JUNC; B 345 PLANO ; B 345 TRIP BRANCH FROM BUS 270730 TO BUS 270928 CKT 1 / ELEC JUNC; B 345 WOLFS ; B 345 TRIP BRANCH FROM BUS 270928 TO BUS 272794 TO BUS 275334 CKT 1 / WOLFS ; B 345 WOLFS ; B 138 WOLFS ;1C 34.5 DISCONNECT BUS 275239 / ELEC JUNC;2M 138 END
COMED_P1-2_345-L1221__B-S	CONTINGENCY 'COMED_P1-2_345-L1221__B-S' TRIP BRANCH FROM BUS 270716 TO BUS 270928 CKT 1 / DRESDE; B 345 WOLFS; B 345 END
COMED_P1-2_345-L10805_B-S	CONTINGENCY 'COMED_P1-2_345-L10805_B-S' TRIP BRANCH FROM BUS 270810 TO BUS 274702 CKT 1 / LOCKP; B 345 KEND
COMED_P4_112-65-BT4-5__	CONTINGENCY 'COMED_P4_112-65-BT4-5__' TRIP BRANCH FROM BUS 270644 TO BUS 243206 CKT 1 / WILTO; 765 05DUMONT 765 TRIP BRANCH FROM BUS 275233 TO BUS 270644 CKT 1 / WILTO;4M 345 WILTO; 765 TRIP BRANCH FROM BUS 275233 TO BUS 270927 CKT 1 / WILTO;4M 345 WILTO; R 345 TRIP BRANCH FROM BUS 275233 TO BUS 275333 CKT 1 / WILTO;4M 345 WILTO;4C 33 END
COMED_P7_345-L0301__B-S_+_345-L9801__R-S_FSA	CONTINGENCY 'COMED_P7_345-L0301__B-S_+_345-L9801__R-S_FSA' TRIP BRANCH FROM BUS 270854 TO BUS 270790 CKT 1 / POWER; B 345 KATYDID TRIP BRANCH FROM BUS 918050 TO BUS 270769 CKT 1 / AA1-018 GOODINGS ;2R 345 END
COMED_P4_111-45-L16704T	CONTINGENCY 'COMED_P4_111-45-L16704T' TRIP BRANCH FROM BUS 270846 TO BUS 270730 CKT 1 / PLANO ; B 345 ELEC JUNC; B 345 TRIP BRANCH FROM BUS 272250 TO BUS 272278 CKT 1 / PLANO ; B 138 PLANO;1I 138 TRIP BRANCH FROM BUS 272250 TO BUS 272278 CKT 2 / PLANO ; B 138 PLANO;1I 138 TRIP BRANCH FROM BUS 270846 TO BUS 272278 TO BUS 275354 CKT 1 / PLANO ; B 345 PLANO;1I 138 PLANO;1C 34.5 TRIP BRANCH FROM BUS 270730 TO BUS 270916 CKT 1 / ELEC JUNC; B 345 WAYNE ; B 345 TRIP BRANCH FROM BUS 270730 TO BUS 270928 CKT 1 / ELEC JUNC; B 345 WOLFS ; B 345 TRIP BRANCH FROM BUS 270928 TO BUS 272794 TO BUS 275334 CKT 1 / WOLFS ; B 345 WOLFS ; B 138 WOLFS ;1C 34.5 DISCONNECT BUS 275239 / ELEC JUNC;2M 138 END

Contingency Name	Contingency Definition
COMED_P1-2_345-L0304__R-S	CONTINGENCY 'COMED_P1-2_345-L0304__R-S' TRIP BRANCH FROM BUS 270855 TO BUS 349662 CKT 1 / POWER; R 345 7TAZEWELL 345 END
COMED_P7_345-L11620_B-S+_345-L11622_R-S	CONTINGENCY 'COMED_P7_345-L11620_B-S+_345-L11622_R-S' TRIP BRANCH FROM BUS 270736 TO BUS 270770 CKT 1 / ELWOO; B 345 GOODI;3B 345 TRIP BRANCH FROM BUS 270737 TO BUS 270769 CKT 1 / ELWOO; R 345 GOODI;1R 345 END
COMED_P7_345-L19601_B-S+_345-L9801__R-S_FSA	CONTINGENCY 'COMED_P7_345-L19601_B-S+_345-L9801__R-S_FSA' TRIP BRANCH FROM BUS 270790 TO BUS 270770 CKT 1 / KATYD; B 345 GOODI;4B 345 TRIP BRANCH FROM BUS 918050 TO BUS 270769 CKT 1 / AA1-018 GOODINGS ;2R 345 END
COMED_P1-2_345-L1202__B-S-A	CONTINGENCY 'COMED_P1-2_345-L1202__B-S-A' TRIP BRANCH FROM BUS 270716 TO BUS 930750 CKT 1 / DRESDEN ; B 345 AB1-122 TAP 345 END
COMED_P4_112-65-BT2-3__	CONTINGENCY 'COMED_P4_112-65-BT2-3__' TRIP BRANCH FROM BUS 270644 TO BUS 270607 CKT 1 / WILTO; 765 COLLI; 765 TRIP BRANCH FROM BUS 275232 TO BUS 270644 CKT 1 / WILTO;3M 345 WILTO; 765 TRIP BRANCH FROM BUS 275232 TO BUS 270926 CKT 1 / WILTO;3M 345 WILTO; B 345 TRIP BRANCH FROM BUS 275232 TO BUS 275332 CKT 1 / WILTO;3M 345 WILTO;3C 33 END
COMED_P4_112-65-BT3-4__	CONTINGENCY 'COMED_P4_112-65-BT3-4__' TRIP BRANCH FROM BUS 270644 TO BUS 243206 CKT 1 / WILTO; 765 05DUMONT 765 TRIP BRANCH FROM BUS 275232 TO BUS 270644 CKT 1 / WILTO;3M 345 WILTO; 765 TRIP BRANCH FROM BUS 275232 TO BUS 270926 CKT 1 / WILTO;3M 345 WILTO; B 345 TRIP BRANCH FROM BUS 275232 TO BUS 275332 CKT 1 / WILTO;3M 345 WILTO;3C 33 END
COMED_P4_012-45-BT5-6__	CONTINGENCY 'COMED_P4_012-45-BT5-6__' TRIP BRANCH FROM BUS 270716 TO BUS 270736 CKT 1 / DRESD; B 345 ELWOO; B 345 TRIP BRANCH FROM BUS 270736 TO BUS 270737 CKT 1 / ELWOO; B 345 ELWOO; R 345 TRIP BRANCH FROM BUS 274702 TO BUS 270716 CKT 1 / KEND
Base Case	
COMED_P4_023-65-BT2-3__	CONTINGENCY 'COMED_P4_023-65-BT2-3__' TRIP BRANCH FROM BUS 270644 TO BUS 243206 CKT 1 / WILTO; 765 05DUMONT 765 TRIP BRANCH FROM BUS 270607 TO BUS 270630 CKT 1 / COLLI; 765 PLANO; 765 END

Contingency Name	Contingency Definition
COMED_P2-2_111_EJ-345B__2	CONTINGENCY 'COMED_P2-2_111_EJ-345B__2' TRIP BRANCH FROM BUS 270730 TO BUS 270846 CKT 1 / ELECT; B 345 PLANO; B 345 TRIP BRANCH FROM BUS 270730 TO BUS 270916 CKT 1 / ELECT; B 345 WAYNE; B 345 TRIP BRANCH FROM BUS 270730 TO BUS 270928 CKT 1 / ELECT; B 345 WOLFS; B 345 TRIP BRANCH FROM BUS 270928 TO BUS 272794 TO BUS 275334 CKT 1 / WOLFS; B 345 WOLFS; B 138 WOLFS;1C 34.5 DISCONNECT BUS 275239 / ELECT;2M 138 END
COMED_P4_112-65-BT5-6__	CONTINGENCY 'COMED_P4_112-65-BT5-6__' TRIP BRANCH FROM BUS 270644 TO BUS 270607 CKT 1 / WILTO; 765 COLLI; 765 TRIP BRANCH FROM BUS 275233 TO BUS 270644 CKT 1 / WILTO;4M 345 WILTO; 765 TRIP BRANCH FROM BUS 275233 TO BUS 270927 CKT 1 / WILTO;4M 345 WILTO; R 345 TRIP BRANCH FROM BUS 275233 TO BUS 275333 CKT 1 / WILTO;4M 345 WILTO;4C 33 END
AEP_P1-2_#695A	CONTINGENCY 'AEP_P1-2_#695A' OPEN BRANCH FROM BUS 243206 TO BUS 270644 CKT 1 / 243206 05DUMONT 765 270644 WILTON ; 765 1 END
AEP_P4_#2978_05DUMONT 765_B	CONTINGENCY 'AEP_P4_#2978_05DUMONT 765_B' OPEN BRANCH FROM BUS 243206 TO BUS 243207 CKT 1 / 243206 05DUMONT 765 243207 05GRNTWN 765 1 OPEN BRANCH FROM BUS 243206 TO BUS 270644 CKT 1 / 243206 05DUMONT 765 270644 WILTON ; 765 1 END
COMED_P4_023-65-BT4-5__	CONTINGENCY 'COMED_P4_023-65-BT4-5__' TRIP BRANCH FROM BUS 275168 TO BUS 270607 CKT 1 / COLLI;2M 345 COLLI; 765 TRIP BRANCH FROM BUS 275168 TO BUS 270697 CKT 1 / COLLI;2M 345 COLLI; R 345 TRIP BRANCH FROM BUS 275168 TO BUS 275268 CKT 1 / COLLI;2M 345 COLLI;2C 33 TRIP BRANCH FROM BUS 270644 TO BUS 243206 CKT 1 / WILTO; 765 05DUMONT 765 END

12 Light Load Analysis

12.1 Generation Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

None

12.2 Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

None

12.3 Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

None

12.4 Potential Congestion due to Local Energy Deliverability

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Note: Only the most severely overloaded conditions are listed below. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed which shall study all overload conditions associated with the overloaded element(s) identified.

None

13 Affected Systems

13.1 TVA

None

13.2 Duke Energy Progress

None

13.3 MISO

MISO Impacts to be determined during later study phases (as applicable).

13.4 LG&E

None

14 Short Circuit

The following Breakers are overduty:

None.

15 Stability and Reactive Power Requirement

Stability Analysis will be performed during the Facilities Study. **Additional reinforcements may be identified to mitigate stability concerns.**

16 Attachment 1 – Single Line Diagram

